

# Session 121: 5G/6G Signal Processing and ISAC



SPCOM 2026

International Conference on  
Signal Processing and Communications

TIME 2:30 PM – 4:00 PM IST VENUE SP 1.20 (TVS Lab)

121.1

## Low Complexity FPGA Implementation of 2×2 MIMO-OTSM Transceiver for 6G Communications

Sapta Girish Neelam

121.2

## Bayesian Multi-Snapshot Learning for Sparse Doubly-Selective CSI Estimation in DCO-OTFS VLC Systems

Shubham Saxena, Suraj Srivastava, Aditya Jagannatham

121.3

## Joint Recovery of Two Carriers from One-bit Sampled Signals

Aparna Agrawal, Kumar Appaiah, Sibi Raj B Pillai

121.4

## Robust URLLC Link Adaptation via Statistical Modeling of Time-Varying SINR Deviations

Shubham Baranwal, Manav Bhatnagar

121.5

## Real-Time Ephemeris Update Mechanism for Effective Doppler and Timing Pre-Compensation in 5G NR-NTN gNB

Akula Chinna Veeresh

121.6

## DMRS-Driven Deep Learning Network for Doppler Estimation in 5G NR-NTN

Ipsita Sannigrahi, Sathish Nallamolu, Pradeep Goutam, Naru Pradeep

121.7

## Differential Cross-Attention based Network for Robust Modulation Classification in OTFS Systems

Sanjeev Sharma, Kuntal Deka

121.8

## RCoD-AoANet: Residual Covariance Denoising for AoA Estimation in ISAC-OTFS Vehicular Systems

Kanwardeep Singh Gahlot, Sandeep Joshi

121.9

## Learning-Based Delay, Doppler, and Angle Estimation Using Quantized DDAE for OFDM Bistatic ISAC

Shiv Shankar

121.10

## On the Joint Impact of UAV Orientation Instability and Phase Errors in UAV-Aided RIS Systems

Sourosmita Das, Manojkumar Kokare, Prashant Sharma, Swaminathan Ramabadran, Sumit Gautam, Saptarshi Ghosh

121.11

## SRS-Based CSI Quality Assessment and Selection for MU-MIMO Precoding in 5G NR

Lalit Bharti, Adarsh Ravi, Preetam Kumar

121.12

## Dual-Stream CNN--LSTM Fusion for Pilot-Assisted Automatic Modulation Classification in OTFS Systems

Kamal Captain, Arvind Chandak

121.13

## Cross-Generation TDD Sensing for 6G ISAC: A Feasibility Study with Legacy LTE and NR Infrastructure

Anindya Saha

121.14

## SNF-PRP: A Covert Integrating Sensing and Communications Framework

Dhruvil Bhatt, Vidushi Kumar

121.15

## Low-Latency Deep Learning Based Doppler Velocity Estimation for Integrated Sensing and Communication

Samarth Sharma Bhardwaj, Aakanksha Tewari, Syed Asrar ul haq, Sumit Darak

# Session 122: Multimodal Learning and Prediction



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TIME 2:30 PM – 4:00 PM IST VENUE SP Building Side Foyer

122.1

**AgentBook: Enabling Fine-Grained Editability in Visual Narratives via Page-Centric Agent Coordination**

Ayushman Sarkar

122.2

**Spectral Decoupling in Attention Heads: Tracking the Onset of Memorization in Vision Transformers**

Dibyajyoti Jena, Sumantra Dutta Roy, Vinayak Abrol

122.3

**Do Latent Representations of Deep Visual Architectures Follow the Population Manifold Hypothesis of the Primate AIT Cortex?**

Ankit Sharma, Sumohana Channappayya

122.4

**AVT-PAC: A Pipeline For Multimodal Action Prediction and Captioning**

Athira Krishnan R, Ambarish Parthasarathy, Swapna M, Sumohana Channappayya

122.5

**Character-aware Subtitling through Improved Alignment and Diverse Priors**

Adhiraj Deshmukh, Makarand Tapaswi, Vineet Gandhi

122.6

**Task-Adaptive FFN Editing for Continual Blind Image Quality Assessment**

Satish Maurya, Parimala Kancharla

122.7

**EchoVision: A Decoupled Audio-Visual Retrieval Framework for Video Question Answering**

Subhajit Sarkar, Joyita Chakraborty, Suman Samui, Subrata Nandi

122.8

**Towards an Efficient and Unified Strategy for Video Understanding Applications**

Rahul Kumar, Sumohana Channappayya

122.9

**Bidirectional Convolutional-GRU Integrated GAN for Unpaired Low-Light Image Enhancement**

Santosh Panda, Pankaj Kumar Sa

122.10

**Getting Uncertainty Right: Local Calibration for Wind Power Forecasting**

Manush Sanchela, Vrishin Rajpurohit, Pritam Anand

122.11

**Microstate temporal dynamics predict valence but not arousal in tactile stimulation**

Suranjita Ganguly, Mohan Raghavan

122.12

**VGGT-Based COLMAP-Free Initialization for Sparse-View 3D Gaussian Splatting**

Shreeya Venkatraman, Yuvan Raj Krishna, Jiji C V

# Session 123: Biomedical Vision and Sensing

TIME 2:30 PM – 4:00 PM IST VENUE ECE 1.07



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## 123.1 A Slice-Based Deep Network for Brain Age Prediction From MRI

Sanjay Ghosh

## 123.2 cAAE-EchoGAN: A Conditional Adversarial Autoencoder Framework for Joint Echocardiographic Image and Segmentation Mask Generation

Shivani Singh, Ashutosh Singh, Shanti Chandra

## 123.3 Scale-aware Economy-Accuracy Trade-off in Contactless Palmprint Recognition Using 14-Tap Q-Shift DTCWT and Layer-Wise Attention

Divyang Jadav, Inderjeet, Hemantkumar Goklani, Vikram M Gadre

## 123.4 DFANet: Dual-Branch Feature Fusion with Multi-Attention Decoding for Colonoscopy Polyp Segmentation

Srinija Kokkanti, Srikanth Panigrahi

## 123.5 Swin-DSA-Net: A Dual-Stage Attention Guided Swin Transformer Network with Multi-scale Refinement for Skin Lesion Segmentation

Nitin Yadav, Bibek Goswami, M Bhuyan

## 123.6 A 3D Channel-Guided Layer Attention Network for Layer-Specific Structural Biomarker Analysis in Optical Coherence Tomography

Mane Pooja, Manish Bhatt, Samarendra Dandapat

## 123.7 SmartVitals-BP: Contactless Blood Pressure Estimation from Facial Videos via Multi-Scale Temporal Modeling and Metadata Fusion

Srinivas Routhu

## 123.8 SwinNetraGF: A Gated Attention Fusion Transformer for Retinal Disease Classification

Anish Vishwakarma, Preity, Rakesh Ranjan

## 123.9 A Multiscale Wavelet-Graph Spectral Approach for Cardiovascular Risk Assessment Using Retinal Fundus Images

S Shrivatssan, Malaya Nath